

**Mathematical Modelling of Biological Dynamics:
Bacterial Resistance to Multiple Antibiotics
and Logistic Population Harvesting**

A Project

**Submitted in partial fulfillment of the Award of the Degree of
B.Sc. (Hons.) in Mathematics**

By

Group-17

ID: B200302024

Session: 2020-2021



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11 May,2026

Abstract

This project presents mathematical models to study biological systems involving bacterial infection dynamics and population growth under harvesting. Systems of ordinary differential equations are used to describe interactions between sensitive and resistant bacteria, immune response, and logistic population growth with both constant and periodic harvesting. Stability analysis identifies conditions for persistence or elimination of populations, revealing that resistant bacteria may survive treatment and that excessive harvesting can lead to population extinction. Periodic harvesting introduces time-dependent behavior that can improve sustainability when properly managed. Numerical simulations support the analytical results. Overall, the study highlights the importance of balanced parameters and demonstrates how mathematical modelling can provide valuable insights for biological control and resource management.

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